

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 1

MH1100 Calculus I

Tutorial 7, Week 8

Your tutor will aim to discuss: Problem 1, 2, 3, and Midterm Test Paper

Problem 1 Find the derivatives of the functions below.

(a) $y = \cot x$

(b) $y = \sec x$

(c) $y = \csc x$

[Solution:]

(a)

$$\begin{aligned} y'(x) &= \left(\frac{\cos x}{\sin x} \right)' = \frac{(\cos x)' \sin x - \cos x (\sin x)'}{\sin^2 x} \\ &= \frac{-\sin x \sin x - \cos x \cos x}{\sin^2 x} \\ &= \frac{-1}{\sin^2 x} = -\csc^2 x. \end{aligned}$$

(b)

$$\begin{aligned} y'(x) &= \left(\frac{1}{\cos x} \right)' = \frac{(1)' \cos x - 1(\cos x)'}{\cos^2 x} \\ &= \frac{\sin x}{\cos^2 x} = \tan x \sec x. \end{aligned}$$

(c)

$$\begin{aligned} y'(x) &= \left(\frac{1}{\sin x} \right)' = \frac{(1)' \sin x - 1(\sin x)'}{\sin^2 x} \\ &= \frac{-\cos x}{\sin^2 x} = -\cot x \csc x. \end{aligned}$$

Problem 2

- (a) The curve $y = |x|/\sqrt{2-x^2}$ is called a *bullet-nose curve*. Find an equation of the tangent line to this curve at the point $(1, 1)$.
- (b) Illustrate part (a) by graphing the curve and the tangent line on the same screen.

[Solution:]

- (a) We want to find an equation of the tangent line to this curve at the point $(1, 1)$. The derivative of the function y at $x = 1$ is needed. When $x > 0$, $y = |x|/\sqrt{2-x^2} =$

$x/\sqrt{2-x^2}$. Its derivative can be evaluated as

$$\begin{aligned}
 y'(x) &= \frac{\sqrt{2-x^2}(x)' - x(\sqrt{2-x^2})'}{(\sqrt{2-x^2})^2} \\
 &= \frac{\sqrt{2-x^2} - x\left(\frac{(2-x^2)'}{2\sqrt{2-x^2}}\right)}{2-x^2} \\
 &= \frac{\sqrt{2-x^2} - x\left(\frac{-2x}{2\sqrt{2-x^2}}\right)}{2-x^2} \\
 &= \frac{\frac{(2-x^2)+x^2}{\sqrt{2-x^2}}}{2-x^2} \\
 &= 2(2-x^2)^{-\frac{3}{2}}.
 \end{aligned}$$

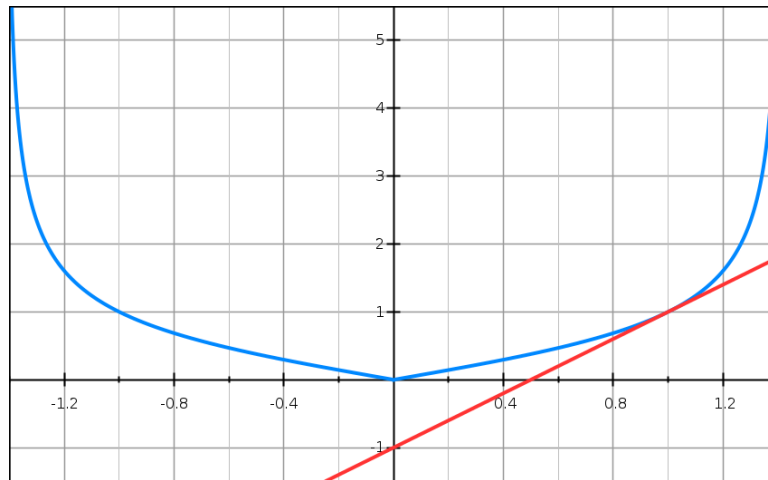
The slope at $(1, 1)$ is $m = y'|_{x=1} = 2$. Using the slope-point equation, we can get the equation of the tangent line to the curve at the point $(1, 1)$ as

$$y - 1 = 2(x - 1).$$

Or

$$y = 2x - 1.$$

- (b) The graphs of y and the tangent line (red) to the curve at the point $(1, 1)$ are shown in the below figure.



Problem 3

The derivative function $f'(x)$ of a differentiable function $f(x)$ is not necessary to be continuous.

Consider the function

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

- (a) Show that $f(x)$ is continuous in its domain.

(b) Show that $f(x)$ is differentiable everywhere.

(c) Show that $f'(x)$ is not a continuous function.

[Solution:]

- (i) When $x \neq 0$, $f(x)$ is obviously continuous. At $x = 0$, using the squeeze theorem by selecting $g(x) = -x^2$ and $h(x) = x^2$ with $g(x) \leq f(x) \leq h(x)$, we can find that

$$0 = \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} h(x) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right).$$

As $f(0) = 0$, the function $f(x)$ is continuous at $x = 0$. So, $f(x)$ is continuous in its domain $\mathbb{R}$.

- (ii) When $x \neq 0$,

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) + x^2 \cos\left(\frac{1}{x}\right) \cdot \left(\frac{1}{x}\right)' = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right).$$

At the point $x = 0$

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 \sin\left(\frac{1}{h}\right) - 0}{h} \\ &= \lim_{h \rightarrow 0} \left[h \sin\left(\frac{1}{h}\right) \right] \\ &= 0. \end{aligned}$$

So $f(x)$ is differentiable everywhere.

- (iii) When x approaches 0, $2x \sin\left(\frac{1}{x}\right)$ approaches 0, but $-\cos\left(\frac{1}{x}\right)$ oscillates more and more frequently. So $\lim_{x \rightarrow 0} f'(x)$ does not exist. $f'(x)$ is therefore not continuous at $x = 0$.